Exercise 2

IAD2133 | OBJECT ORIENTED PROGRAMMING
Chapter 2 - Elementary Programming

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1. Scanner Class

Which method in the `Scanner` class is used specifically to read a value of type `double` from the console?

A. `nextDouble()`

B. `getDouble()`

C. `inputDouble()`

D. `readDouble()`

---

2. Java Identifiers (True/False)

A Java identifier can legally begin with a digit, provided the rest of the characters are letters or underscores.

A. True B. False

---

3. Valid Identifier

Which of the following is a valid Java identifier?

A. 123geeks

B. sum_&_difference

C. My Variable

D. $myvariable

---

4. Modulus Operation

If you calculate (6 + 10) % 7, what is the result?

A. 1

B. 3

C. 2

D. 0

---

5. Floating-Point Literal

By default, a floating-point literal such as 5.0 is treated as which data type in Java?

A. float

B. long

C. double

D. int

---

6. Integer Division

What is the result of `5 / 2` in Java?

A. 2.5
B. 3
C. 2
D. 0

---

7. System Time

The method `System.currentTimeMillis()` returns the number of milliseconds elapsed since:

A. The time the program started
B. Midnight today
C. January 1, 1900 GMT
D. January 1, 1970 GMT

---

8. Type Casting

When casting a `double` to an `int`, such as `(int) 3.9`, what happens?

A. Fractional part is truncated
B. Stored elsewhere
C. Runtime error
D. Rounded

---

9. Software Development Phase

Which phase focuses on understanding the problem and documenting system requirements?

A. System Analysis

B. Requirements Specification

C. Implementation

D. System Design

---

10. Type Conversion

If one operand is `double` and the other is `int`, what is the result type?

A. float

B. long

C. int

D. double

---

11. System Analysis (True/False)

System Analysis involves identifying classes, interfaces, and relationships.

A. True

B. False

12. Augmented Assignment

Which is an example of an augmented assignment operator?

A. `+=`

B. `&&`

C. `++`

D. `==`

13. Acronym in System Design

What does IPO stand for?

A. Input, Process, Output

B. Application Programming Interface

C. Graphical User Interface

D. Integrated Development Environment

14. Expression Equivalence (True/False)

If `sum` is an integer, then `sum += 4.5` is equivalent to `sum = (int)(sum + 4.5)`.

A. True

B. False

15. Data Type Error

What error occurs when a value exceeds a data type's range?

A. Undeclared Variable

B. Integer Overflow

C. Round-off Error

D. Redundant Input

---

16. Maintenance Phase

Which phase focuses on improving the system and fixing bugs over time?

A. Implementation

B. Maintenance

C. Deployment

D. Testing

---

17. Expression Evaluation

What is the value of:

```
3 + 4 * 4 + 5 * (4 + 3) - 1
```

A. 83

B. 53

C. 112

D. 48

18. Identifier Length (True/False)

Identifiers in Java can be of any length.

A. True

B. False

---

19. Character Data Type

Which data type stores a single character in Java?

A. char

B. boolean

C. byte

D. String

20. Return Type (True/False)

`currentTimeMillis()` returns a value of type `int`.

A. True

B. False

Exercise code 1: Area of a Circle

Concept: Variables, Constants, and Basic Input.

Problem: Write a program that reads the radius of a circle from the user and computes the area. Use the constant Math.PI for the calculation.

- **Formula:** $Area = radius \times radius \times \pi$

```java
import java.util.Scanner;
class AreaOfCircle{

    public static void main(String[] args){

    Scanner input = new Scanner(System.in);
        System.out.println("Enter radius: ");
        double radius = input.nextDouble();

        double area = radius * radius * Math.PI;

        System.out.println("The area of the circle is " + area);
    }
}
```

```
PS C:\Users\azfar> cd desktop
PS C:\Users\azfar\desktop> cd oop
PS C:\Users\azfar\desktop\oop> java AreaOfCircle
Enter radius:
15
The area of the circle is 706.8583470577034
```

Exercise code 2: Temperature Converter

Concept: Arithmetic Expressions and Floating-Point Math.

Problem: Write a program that reads a Celsius degree in a double value from the console, then converts it to Fahrenheit and displays the result.

- **Formula:** $Fahrenheit = (9 / 5) \times Celsius + 32$ ● *Hint: Be careful with integer division when calculating 9/5!*

```java
import java.util.Scanner;

public class TempConverter{
    public static void main(String[] args){

        Scanner input = new Scanner(System.in);
        System.out.println("Enter Celcius: ");

        double celcius = input.nextDouble();
        double fahrenheit = (9/5) * celcius + 32;

        System.out.println(celcius + " Celcius is " + fahrenheit + " Fahrenheit");

    }
}
```

```
PS C:\Users\azfar> cd desktop
PS C:\Users\azfar\desktop> cd oop
PS C:\Users\azfar\desktop\oop> java TempConverter
Enter Celcius:
25
25.0 Celcius is 57.0 Fahrenheit
PS C:\Users\azfar\desktop\oop>
```

Exercise code 3: Time Decomposition

Concept: Integer Division (/) and Remainder Operator (%).

Problem: Write a program that prompts the user for a total number of seconds (e.g., 500) and converts it into minutes and remaining seconds (e.g., 8 minutes and 20 seconds).

```java
public class TimeDecomposition{

    public static void main(String[] args){

        Scanner input = new Scanner(System.in);
        System.out.println("Enter number of seconds: ");

        int number = input.nextInt();
        int minutes = number / 60;
        int remainingSecs = number % 60;

        System.out.println(number + " seconds is " + minutes + " minutes and " + remainingSecs + " seconds.");
    }
}
```

```
PS C:\Users\azfar> cd desktop
PS C:\Users\azfar\desktop> cd oop
PS C:\Users\azfar\desktop\oop> java TimeDecomposition
Enter number of seconds:
500
500 seconds is 8 minutes and 20 seconds.
PS C:\Users\azfar\desktop\oop>
```

Exercise code 4: Digit Summation

Concept: Modulo and Division for digit extraction.

Problem: Write a program that reads an integer between 0 and 1000 and adds all the digits in the integer. For example, if an integer is 932, the sum of all its digits is $9 + 3 + 2 = 14$. Java

```java
import java.util.Scanner;

public class DigitSummation{

    public static void main(String[] args){

        Scanner input = new Scanner(System.in);
        System.out.println("Enter a number between 0 and 1000: ");

        int number = input.nextInt();
        int LastDigit = number % 10;
        int Remaining = number / 10;
        int SecondDigit = Remaining % 10;
        int FirstDigit = Remaining / 10;
        int sum = FirstDigit + SecondDigit + LastDigit;

        System.out.println("The sum of all digits is: " + sum);
    }
}
```

```
PS C:\Users\azfar> cd desktop
PS C:\Users\azfar\desktop> cd oop
PS C:\Users\azfar\desktop\oop> java DigitSummation
Enter a number between 0 and 1000:
120
The sum of all digits is: 3
```

Exercise code 5: Gratuity Calculator

Concept: Combining multiple inputs and percentages.

Problem: Write a program that reads the subtotal and the gratuity rate (percentage), then computes the gratuity and the total. For example, if the user enters 10 for subtotal and 15% for gratuity rate, the program displays $1.5 as gratuity and $11.5 as total.

```java
import java.util.Scanner;

public class GratuityCalc{
    public static void main(String[] args){

        Scanner input = new Scanner(System.in);
        System.out.println("Enter subtotal: ");

        int subtotal = input.nextInt();

        System.out.println("Enter gratuity rate: ");

        double rate = input.nextDouble();

        double gratuity = subtotal * (rate / 100);
        double total = subtotal + gratuity;

        System.out.println(gratuity + " gratuity and " + total + " total.");
    }
}
```

```
PS C:\Users\azfar> cd desktop
PS C:\Users\azfar\desktop> cd oop
PS C:\Users\azfar\desktop\oop> java GratuityCalc
Enter subtotal:
15
Enter gratuity rate:
10
1.5 gratuity and 16.5 total.
PS C:\Users\azfar\desktop\oop>
```

Exercise code 6: Exponentiation and Math Functions

Concept: Using the Math.pow(a, b) method.

Problem: Write a program that displays a table of powers. For each number n from 1 to 5, calculate n^{n+1}.

- *Expected output row 1:* 1 2 1 (where 1 is n, 2 is $n+1$, and 1 is 1^2) Java

```java
public class power{
    public static void main(String[] args){

        System.out.println("n^{n+1}");
        for (int n=1;n<=5;n++){
            int results = (int) Math.pow (n,n+1);

            System.out.println(n + " is " + results);
        }
    }
}
```

```
n^{n+1}
1 is 1
2 is 8
3 is 81
4 is 1024
5 is 15625
[Finished in 1.8s]
```

Exercise code 7: Numeric Type Casting

Concept: Implicit vs. Explicit Casting.

Problem: Create a program that prompts the user for three integers, calculates their average, and displays it as a double. Note: Ensure the result has decimal precision (e.g., the average of 1, 2, and 4 should be 2.333, not 2.0).

```java
import java.util.Scanner;

public class Average{
    public static void main(String [] args){

        Scanner input = new Scanner(System.in);
        System.out.println("Enter three integers: ");

        int number1 = input.nextInt();
        int number2 = input.nextInt();
        int number3 = input.nextInt();
        double average = (number1 + number2 + number3) / 3.0;

        System.out.println("The result for average is " + average);
    }
}
```

```
PS C:\Users\azfar> cd desktop
PS C:\Users\azfar\desktop> cd oop
PS C:\Users\azfar\desktop\oop> java Average
Enter three integers:
1
2
3
The result for average is 2.0
```